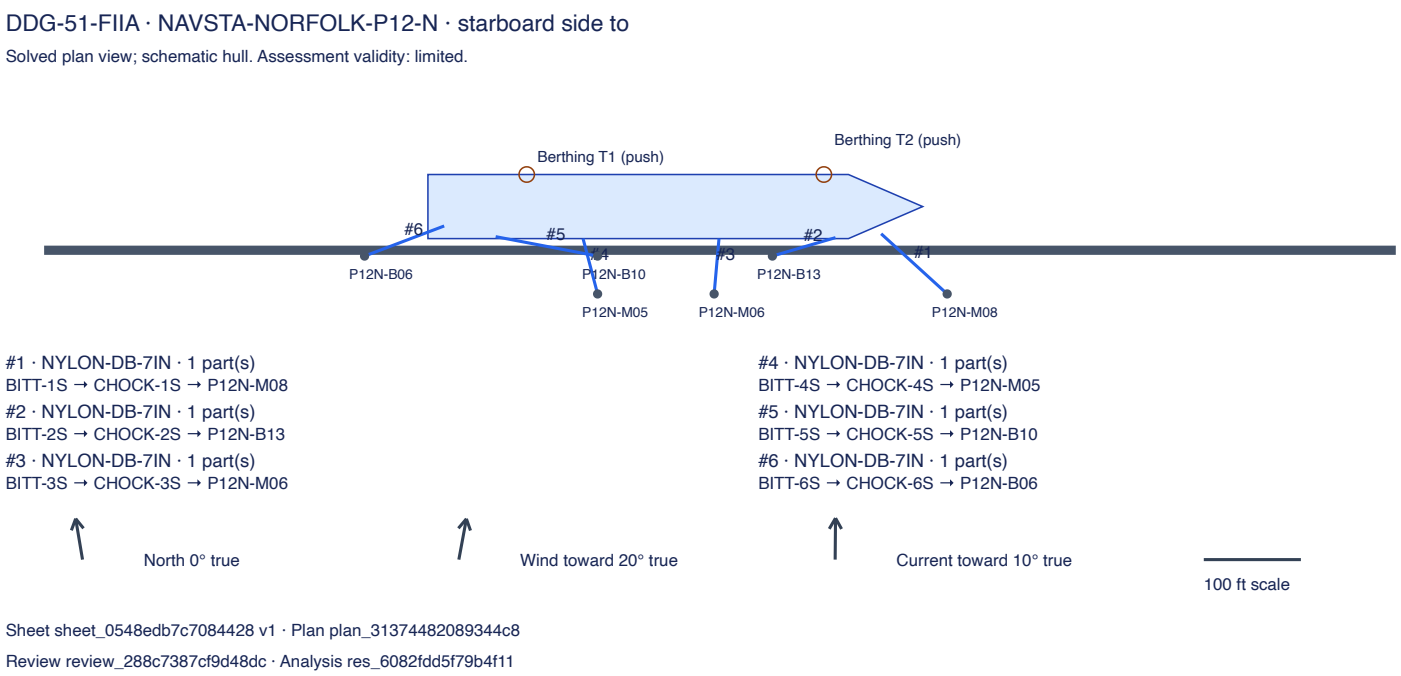
Mooring Evolution Safety Sheet

Validity: limited. Seed or estimated reference data is not qualified for operational use.

Sheet sheet_0548edb7c7084428 v1 · Status **AMBER** · DDG-51-FIIA starboard side to NAVSTA-NORFOLK-P12-N

Independent review: review_288c7387cf9d48dc.

Assessed graphical layout



Wind 4 kt gusting 5 kt from 200° true; current 0.1 kt flowing toward 010° true; water level 1.80 ft MLLW; assessment MST II

Evaluated condition: 2026-11-05 18:46 +0000; source scenario. These are assessed inputs, not a live observation display.

Line-installation water level: 1.8 ft MLLW; held fixed across assessed conditions, with no implicit tending.

Wind averaging, gust duration and reference height: not recorded in these inputs.

Forecast assessment

No windows of concern; MST required is I; it does not exceed the ship's MST II. Next 48 h at Pier 12 North face (NAVSTA Norfolk): 0 window(s) of concern. Gusts peak 5 kt; MST I conditions.

Directional gust thresholds

Modeling result references: res_6a56d881a32343d1.

Relative wind	Governing element	Working-load limit	Breaking threshold estimate
0°	line-5	100 kt	Unavailable; not established

Relative wind	Governing element	Working-load limit	Breaking threshold estimate
15°	line-1	70 kt	Unavailable; not established
30°	line-3	52 kt	Unavailable; not established
45°	line-3	43 kt	Unavailable; not established
60°	line-3	40 kt	Unavailable; not established
75°	line-3	40 kt	Unavailable; not established
90°	line-3	41 kt	Unavailable; not established
105°	line-3	42 kt	Unavailable; not established
120°	line-3	43 kt	Unavailable; not established
135°	line-3	49 kt	Unavailable; not established
150°	line-4	60 kt	Unavailable; not established
165°	line-6	83 kt	Unavailable; not established
180°	line-2	87 kt	Unavailable; not established
195°	line-2	84 kt	Unavailable; not established
210°	line-2	88 kt	Unavailable; not established
225°	line-2	116 kt	Unavailable; not established
240°	P12N-CAMEL-1	120 kt	Unavailable; not established
255°	P12N-CAMEL-1	123 kt	Unavailable; not established
270°	P12N-CAMEL-2	86 kt	Unavailable; not established
285°	P12N-CAMEL-2	67 kt	Unavailable; not established
300°	line-4	57 kt	Unavailable; not established
315°	line-4	55 kt	Unavailable; not established
330°	line-4	58 kt	Unavailable; not established
345°	line-4	70 kt	Unavailable; not established

Line plan

Static loads for the evaluated condition, result res_6082fdd5f79b4f11. Other reviewed conditions may govern individual elements.

#	Role	Line	Parts	Ship fitting	Pier fitting	Tension (lbf)	% WLL	Status
1	bow	NYLON-DB-7IN	1	BITT-1S	P12N-M08	2,167	6%	GREEN
2	spring_aft	NYLON-DB-7IN	1	BITT-2S	P12N-B13	2,164	6%	GREEN
3	breast	NYLON-DB-7IN	1	BITT-3S	P12N-M06	2,143	6%	GREEN
4	breast	NYLON-DB-7IN	1	BITT-4S	P12N-M05	2,144	6%	GREEN
5	spring_fwd	NYLON-DB-7IN	1	BITT-5S	P12N-B10	2,176	6%	GREEN

#	Role	Line	Parts	Ship fitting	Pier fitting	Tension (lbf)	% WLL	Status
6	stern	NYLON-DB-7IN	1	BITT-6S	P12N-B06	2,163	6%	GREEN

Pier fitting margins

Fitting	Load (lbf)	% capacity	Status
P12N-M08	2,167	0%	GREEN
P12N-B13	2,164	1%	GREEN
P12N-M06	2,143	0%	GREEN
P12N-M05	2,144	0%	GREEN
P12N-B10	2,176	3%	GREEN
P12N-B06	2,163	1%	GREEN

Holding assistance

No prescribed holding assistance is included in the line assessment.

Berthing tug plan

2 tugs, 9,000 hp, 235,000 lbf bollard pull

Go / no-go

- **Gusts above 32 kt (80% of plan limit)**: Reassess the plan with the bridge and pilot; use only an independently reviewed tending or assistance procedure
- **Gusts above 40 kt (working-load limit)**: Call holding tugs; consider an alternate berth

Sequence

DRAFT, not reviewed by a deck officer or pilot (draft_unreviewed). Static check only: transient loads while lines are handled and tugs maneuver are not evaluated; they need the M11 simulator. Rule table is a draft from NSTM 582-2.4.6 paragraph references that were not re-verified in this repository. Durations are unreviewed planning estimates, not measured line-handling times.

Sequence assessment: `res_b6fe6139cd034c27`.

1. pass line 2 (forecastle): Bridge orders first line over
2. secure pier line 2 (pier): Line 2 eye ashore **Static check: within limits (draft).** worst line-2 at 0% of its WLL limit
3. pass line 5 (fantail): Line 2 on its bollard
4. secure pier line 5 (pier): Line 5 eye ashore **Static check: within limits (draft).** worst line-5 at 0% of its WLL limit
5. pass line 3 (waist): Line 5 on its bollard
6. secure pier line 3 (pier): Line 3 eye ashore **Static check: within limits (draft).** worst line-5 at 0% of its WLL limit
7. hold line 3 (waist): Tugs holding the ship alongside *Keep this line slack-tended; don't take it to power until the tugs ease off.*
8. pass line 4 (waist): Line 3 on its bollard

9. secure pier line 4 (pier): Line 4 eye ashore **Static check: within limits (draft).** worst line-5 at 0% of its WLL limit
10. hold line 4 (waist): Tugs holding the ship alongside *Keep this line slack-tended; don't take it to power until the tugs ease off.*
11. pass line 1 (forecastle): Line 4 on its bollard
12. secure pier line 1 (pier): Line 1 eye ashore **Static check: within limits (draft).** worst line-5 at 0% of its WLL limit
13. pass line 6 (fantail): Line 1 on its bollard
14. secure pier line 6 (pier): Line 6 eye ashore **Static check: within limits (draft).** worst line-5 at 0% of its WLL limit
15. hold line - (bridge): All lines on their bollards; tugs ease off **Static check: within limits (draft).** worst line-3 at 2% of its WLL limit
16. take to power line 2 (forecastle): Bridge orders lines taken to power **Static check: within limits (draft).** worst line-2 at 4% of its WLL limit
17. take to power line 5 (fantail): Bridge orders lines taken to power **Static check: within limits (draft).** worst line-2 at 6% of its WLL limit
18. take to power line 3 (waist): Bridge orders lines taken to power **Static check: within limits (draft).** worst line-5 at 6% of its WLL limit
19. take to power line 4 (waist): Bridge orders lines taken to power **Static check: within limits (draft).** worst line-2 at 6% of its WLL limit
20. take to power line 1 (forecastle): Bridge orders lines taken to power **Static check: within limits (draft).** worst line-2 at 8% of its WLL limit
21. take to power line 6 (fantail): Bridge orders lines taken to power **Static check: within limits (draft).** worst line-5 at 6% of its WLL limit
22. tend line - (bridge): All lines doubled up and tensioned *Equalize: slack the shortest lines about 1 ft; pre-tension lines over 100 ft.*

Warnings

- [AMBER] ship 'DDG-51-FIIA' is estimated data, not published/qualified. (Result validity)
- [AMBER] Ship fitting heights are not corrected for the requested loading. (Result validity)

Assessment assumptions and limits

- Configured model: deterministic-fallback; adjustment mode: deterministic.
- Model text is not used as factual rationale; physics results govern status.
- Line installation is fixed at the first planning condition water level (1.8 ft MLLW); no implicit tending between conditions.
- Forecast-only planning checks the supplied conditions; it does not establish MST design-condition compliance. Intake selects representative snapshots, not every forecast sample; use the forecast-plan endpoint for an arrival/holding interval assessment.
- Quasi-static 3-DOF equilibrium (surge, sway, yaw) of a rigid hull; heave, roll, pitch and dynamics are not modeled.
- Applied environmental loads include wind and current only; wave and wake loads are not evaluated.
- Lines are tension-only; tension = parts x effective MBS x curve(strain) with the catalog stretch curve scaled by the effective (wet/dry) MBS.
- Fenders are unilateral springs (catalog linear stiffness) in contact over a parallel body of +/- half the waterline length.
- Pier-fitting loads are the 3-D vector sum of attached lines compared with the nominal capacity.
- Rest lengths are set at the installation water level and stay fixed; the ship rides the water level.
- Design factor 3.0 (MST II).
- line 1: rest length from default 2% pretension.
- line 2: rest length from default 2% pretension.
- line 3: rest length from default 2% pretension.
- line 4: rest length from default 2% pretension.
- line 5: rest length from default 2% pretension.
- line 6: rest length from default 2% pretension.

- Gust duration not given: assumed 30 s (legacy).
- Wind averaging/gust duration/height not documented in the snapshot: legacy assumption applied (10 m, 60 s sustained, 30 s gust).
- No lines fast: tugs must cancel the whole static wind-and-current wrench (surge, sway, yaw) at the berthing heading.
- Catalog bollard pull (seed/estimated unless published); ahead for pushing, astern for pulling, capped by the push point rating; efficiency factor 1. Not certified ratings. Horsepower is not converted to force.
- Margin 25% applied to the wrench.
- UFC 4-159-03 (2020), Chapter 4; Eq. 4-3 harmonic and symmetric angle folding restored from the earlier UFC edition.
- Wind basis follows supplied metadata; unsupported conversions and any legacy assumptions are listed below.
- Wind shielding scales all wind forces and moments equally; current is unshielded.
- Static wind and current loads only; wave and wake loads are not evaluated.
- Water depth is actual depth at the ship; water level is not added again.
- Current model assumes a wide harbor/channel, level keel, and 15 C salt-water viscosity (Table 4-1); density uses the request.
- Yaw uses UFC Tables 4-5/4-10: DDG and CG use destroyer wind / FFG current; CVN and LHD use carrier wind / CVE 55 current. CG is assumed at midships.
- Ship geometry is estimated: Wind-force magnitudes calibrated to NSTM Table 582-3-1 with areas within 25% of original seeds; peak bearings also match after converting NSTM flow-toward angles to API wind-from angles. Transverse drag class is extensive_superstructure (C=1.02), per UFC Table 4-2 for destroyers. Dimensions approximate from public fact sheets; fittings notional. Superstructure height frozen before area calibration using original hull area/LWL plus original superstructure area/(0.5 LWL), assuming a rectangular half-length superstructure. Midship coefficient 0.83 estimated by analogy to UFC Table 4-8 destroyer. Wetted surface estimated with UFC Eq. 4-18 at full_load and density 1020 kg/m3; other loadings recompute the estimate. Propeller expanded area estimated using Eqs. 4-22/4-23 and Table 4-7, pitch ratio 1 (LHD uses carrier analogy).
- Line installation tide: 1.8 ft MLLW; no implicit tending in these draft static checks.
- Loading full_load; strength design factor uses MST II.
- Intermediate states: only deployed parts are modeled (one until double_up), slack until taken to power; tugs hold until every line is on its bollard. Tug release is checked before further tensioning.
- Static check only: transient loads while lines are handled and tugs maneuver are not evaluated; they need the M11 simulator.
- Rule table is a draft from NSTM 582-2.4.6 paragraph references that were not re-verified in this repository.
- Durations are unreviewed planning estimates, not measured line-handling times.
- 200 instantaneous samples of gust factor, direction and current from model 'synthetic-demo' r1; each solved to equilibrium.
- Failed samples are counted, reported and used to bound the probability.
- Fender exceedance compares energy with rated energy; reported load percentiles are reaction force in lbf.
- No dynamic or snatch amplification is modeled; the dynamic factor, if any, scales wind force once.

Agent rationale

- Started from the optimizer's highest-ranked candidate. [res_06777b0415b6448b]
- Preserved modeling-resolved unstretched line spans for subsequent rechecks; these are not rope-cutting lengths or handling instructions. [res_6082fdd5f79b4f11]
- Checked the final configuration at a required planning condition. [res_6082fdd5f79b4f11]
- Checked the final configuration at a required planning condition. [res_eba9df1b590143e0]
- Gust-risk assessment for a required planning condition. [res_b1b74046b51c4e72]
- Gust-risk assessment for a required planning condition. [res_df55c4b6abdc45f5]
- Berthing tugs are sized for the most demanding required condition. [res_6cb3806fcd744428, res_fadc83d2fb544b6e]

Referenced modeling results

- res_06777b0415b6448b
- res_6082fdd5f79b4f11
- res_6a56d881a32343d1
- res_6cb3806fcd744428
- res_6f4b188240744756
- res_b1b74046b51c4e72
- res_b6fe6139cd034c27
- res_df55c4b6abdc45f5
- res_eba9df1b590143e0
- res_fadc83d2fb544b6e

Approvals

- Pending

Decision aid only. The Commanding Officer and pilot retain authority for the mooring evolution. Values marked mock or estimated are not validated.